

# Slow to Hire, Slow to Fire: Labour Adjustment Frictions in French Viticulture

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## Abstract

To what extent are French wine farms slow to adjust their workforce to demand fluctuations? In this paper, we measure the cost of adjusting variable labour using the 2003–2024 French Farm Accountancy Data Network (FADN) panel. Using a quasi-natural experiment, we document that variable labour barely moves when sales collapse. We identify the same sluggishness on the upside with instrumental variables built from foreign-market variation. A one-standard-deviation increase in demand raises variable-labour hiring by about 27 percent, only half of what a farm would add in the absence of any adjustment costs. A structural model with heterogeneous farm productivity and permanent labour acting as a production buffer translates this elasticity into a per-worker contraction cost. Removing the friction would raise the average farm’s operating profit by about 2% per year (about €140 million annually for the sector). The gain is less about cutting labour in downturns than about the hiring farms forgo in good years for fear of costly future contraction. The cost concentrates on farms in the upper half of the productivity distribution.

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# 1 Introduction

French wine is a sector of persistent sales shocks.<sup>1</sup> Farms can meet these swings by expanding or shrinking the workforce, but that adjustment is costly. The labour-adjustment-cost literature has measured that cost at the plant level in manufacturing, but its magnitude in agriculture, and its incidence across heterogeneous farms, have been hard to measure. Adjustment costs have several sources. They can stem from the specific knowledge a worker builds up on the job, from training and coordination, or from institutional features of the labour market such as severance pay and end-of-contract costs.

We do not identify which of these sources is responsible for the adjustment costs; instead, in this paper, we measure them directly, using the French Farm Accountancy Data Network (FADN) panel of 3,872 wine farms over 2003–2024.<sup>2</sup> Adjustment at the farm level happens primarily on variable labour, defined as the non-permanent component of the workforce, including fixed-term contracts renewed across years and seasonal contracts. Our data on this variable reveal two facts that motivate the analysis. About a quarter of farm-years operate without any variable labour at all. In the other three-quarters, variable labour moves with sales, but only about half as much as it would if farms could adjust their workforce freely.

A quasi-natural experiment motivates the analysis and shows the friction at work. When a climate catastrophe makes a farm’s sales collapse, variable labour barely responds, even as the shortfall persists for years. The same friction shapes hiring when sales boom. A forward-looking farm hires less than demand warrants, knowing that workers added today may be costly to shed tomorrow. To identify this response we need changes in the demand for French wine that the farm did not cause. The weather of a large competitor provides them. The United States is a major wine producer and importer, so a poor season in California ripples through the world wine trade and lifts the demand facing French producers in the markets where they compete. That Californian weather reaches a French farm’s labour only through the demand for its wine, never directly. We use the elasticity estimated by this IV strategy as a central input to a structural model à la Cooper and Haltiwanger (2006). We find that the demand increase raises variable-labour hiring well below what an unconstrained farm would add.

Behind this rigidity is a per-worker cost of contracting variable labour. We re-

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<sup>1</sup>Since 2008, the industry has absorbed the financial crisis, a sharp increase and then decrease in Chinese demand, Brexit and U.S. protectionism, stop-and-go disruptions due to COVID, the Russia–Ukraine war, and a run of adverse harvests.

<sup>2</sup>Survey-weighted; full sample description in Section 3.1.

cover it via indirect inference on an adjustment-cost model with two features that fit French viticulture. First, permanent labour produces output even when no variable labour is hired, so a farm can choose to operate with permanent staff alone in low-revenue years. Second, farms differ in a permanent productivity dimension, which we calibrate from the dispersion of farm fixed effects in the production function. The model matches the IV-identified elasticity and the share of farm-years operating with permanent staff alone. From that model, we find that removing the adjustment cost would raise operating profit by about 2% per year on average across the population, on the order of €140 million annually for the French wine sector, holding input and output prices fixed. The cost concentrates on commercial farms in the middle and upper half of the productivity distribution. Median farms pay for keeping variable labour they would otherwise let drop to zero in low-revenue years, while mid-large commercial farms under-hire when demand is high and bear roughly three-quarters of the aggregate loss.

The structural adjustment-cost framework we apply originates with Cooper and Haltiwanger (2006), Cooper and Willis (2009), and Bloom (2009). These papers estimate the parameters of adjustment-cost models, on capital or labour or both, using plant-level manufacturing panels. While agricultural economics has a long literature on farm labour,<sup>3</sup> the structural adjustment-cost framework we apply has not been brought to that setting. We extend the framework in two ways. First, we apply it at the farm level in agriculture; the climate-driven volatility of farm revenue and the option to operate with permanent staff alone are features the manufacturing applications do not face. Second, we recover the auxiliary moment causally, whereas these papers recover it by OLS. This brings to the adjustment-cost literature the causal-IV-into-structural turn that Berry and Compiani (2023) have recently pursued in industrial organisation. Weather-based instruments have a precedent in Roberts and Schlenker (2013), who exploit a crop’s own yield shocks as a supply shifter to estimate world commodity elasticities. Our design departs from theirs on three counts. The instrument is a foreign competitor’s weather, not the home crop’s; it shifts demand through the trade network rather than domestic supply; and the elasticity it identifies, recovered from differential exposure across French regions, feeds a structural labour model rather than a reduced-form market equilibrium.

The paper is organised as follows. Section 2 introduces the model framework that the empirical strategy targets, together with the quasi-natural experiment that motivates the model. Section 3 describes the data and the instrumental variable

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<sup>3</sup>On farmers’ labour supply, see Sumner (1982) and Rutledge and Mérel (2023); on migration and the farm workforce, Charlton and Taylor (2016) and Clemens et al. (2018); on the family-versus-hired-labour distinction, Benjamin (1992) and LaFave and Thomas (2016).

strategy. Section 4 reports the IV estimates. Section 5 closes the model, runs the indirect inference, and quantifies the operating-profit cost of the labour friction across farm types. Section 6 concludes.

## 2 Model framework

This section reviews the sources of labour adjustment cost, documents the friction in a quasi-natural experiment, and sets up the dynamic model of variable-labour choice.

### 2.1 About adjustment costs

A basic assumption in microeconomic textbooks is that labour adjusts instantaneously to minimise costs. In practice, a long literature beginning with Clark (1923) and Oi (1962) has emphasised that labour is often quasi-fixed (e.g., Bouchet et al., 1989; Moschini, 1989). In viticulture, quasi-fixity is arguably even stronger, because grapevines are perennial, many tasks require plot-specific knowledge, and production under quality schemes adds rigidity through rules on practices and yields. For organic production, these frictions are amplified further through pest management without synthetic inputs, compliance requirements, and organisational adaptation (Carneiro et al., 2024).

The literature distinguishes three broad classes of adjustment cost, each grounded in a distinct micro-foundation.

**Convex adjustment costs.** The dynamic labour-demand tradition of Sargent (1978) models the adjustment cost as quadratic in the change of employment. Internal training, ramp-up, and coordination costs are the canonical micro-foundation. In viticulture, site-specific knowledge is the clearest illustration. Every parcel has its own soil profile, slope, microclimate, and rootstock-clone composition, and a new vineyard worker accumulates this knowledge gradually over seasons. The same gradual ramp-up applies across the wine sector, from vineyard work and treatment timing to cellar and lab procedures, and is sharper still on organic or biodynamic estates, which add mechanical-weeding and biocontrol skills to the standard apprenticeship.

**Non-convex (fixed) adjustment costs.** The non-convex employment-adjustment literature (Hamermesh, 1989; Caballero et al., 1997; Cooper and Willis, 2009) treats

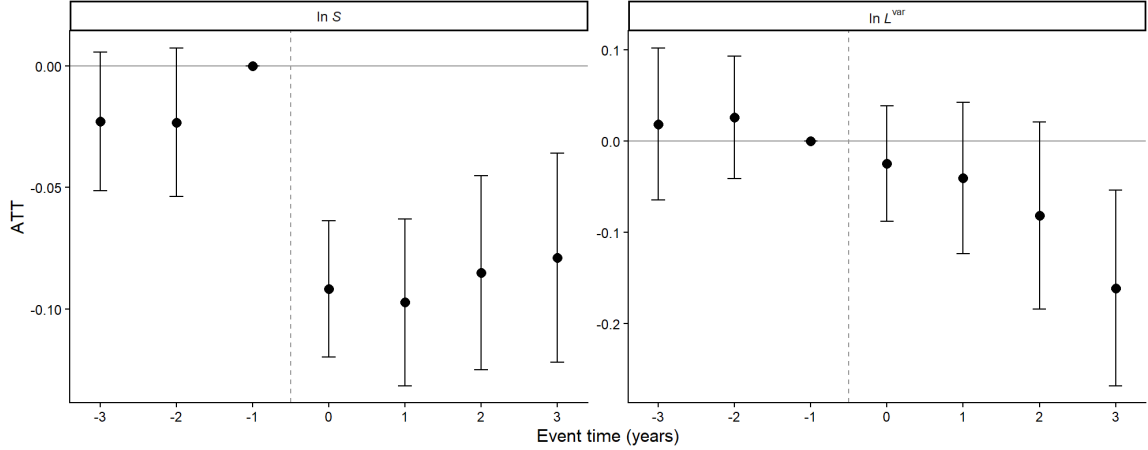
adjustment as a discrete event that triggers a fixed cost. This captures indivisibilities, integer constraints, and contract-negotiation overhead. A vineyard’s pruning team is hired as a unit, not in fractions; a salaried position is either filled or not. Labour-side fixed costs of this kind appear for instance in the per-team recruitment overhead — transport, lodging, paperwork and supervisor setup, paid once per team regardless of headcount. Producers concentrate adjustments into discrete hiring episodes rather than making marginal changes.

**Asymmetric (contraction-only) adjustment costs.** The dynamic labour-demand tradition of Bentolila and Bertola (1990) and Bertola (1992) treats adjustment as costly only on the contraction side. These adjustment costs may be linked to the objective of retaining a qualified workforce through downturns in anticipation of recovery. They can also arise from institutional features of the labour market — severance indemnities, the legal cost of dismissal, end-of-contract bonuses, hiring search frictions — though the asymmetric-contraction-cost concept is generic rather than specific to any one regulatory regime.

The empirical pattern documented in the next subsection points to this asymmetric form, which the structural setup adopts on the variable-labour margin.

## 2.2 A quasi-natural experiment: how variable labour responds to a sharp revenue fall

To illustrate adjustment costs in periods of falling sales, we study how French wine farms respond to a climate catastrophe — a frost, hail, drought, or storm that damages the vineyard or cellar stock. We identify these events from insurance indemnities, recorded at the farm-year level, and treat the year of each farm’s largest payout as the shock, keeping the 918 farms whose largest payout exceeds €10,000. Because the catastrophes strike different farms in different years, we estimate the dynamic response with the interaction-weighted event-study estimator of Sun and Abraham (2021), which is robust to the bias that two-way fixed effects suffer under staggered timing and heterogeneous effects (Goodman-Bacon, 2021). We use farm and year fixed effects and  $h = -1$  as the reference period (construction and robustness in Appendix C). Identification rests on the catastrophe being unforeseeable when the farm commits its labour — so the shock predates the labour choice — and on treated and not-yet-treated farms following parallel paths absent the event, which we verify with the pre-event coefficients. If contracting labour is costly, we expect revenue to fall on impact while variable labour barely adjusts, even as the shortfall persists.



**Note:** Sun–Abraham IW event-study coefficients on log sales ( $\ln S$ , left panel) and log variable labour ( $\ln L^{\text{var}}$ , right panel), by event time  $h = t - g_i$  where  $g_i$  is the year of farm  $i$ 's largest indemnity payout. Cohort restricted to farms with maximum indemnity  $\geq \text{€}10,000$  ( $N = 918$  wine farms). Vertical bars show 95% confidence intervals (farm-clustered). Reference period  $h = -1$ .

Figure 1: Event-study response on the indemnity-amount cohort

The two response paths are revealing (Figure 1). Revenue drops sharply at the event year, by about  $-0.09$  in logs, and stays significantly depressed for three years. Variable labour, by contrast, barely moves. The event-year coefficient, about  $-0.025$ , is statistically indistinguishable from zero, and the response stays small and insignificant through the second post-event year. A larger coefficient of about  $-0.16$  appears at the third year, but it is not robust. As we raise the indemnity threshold that defines the cohort, the estimate shrinks toward zero and loses significance (Appendix C). Pre-event coefficients are jointly insignificant ( $\chi^2(2) = 0.77$ ,  $p = 0.68$ ), supporting parallel pre-trends. The picture is one of a sharp, persistent fall in revenue met with little detectable labour contraction at any horizon.

This asymmetry — a large, persistent revenue shock met with little labour response — is what the next subsection formalises as a cost that bites on the contraction margin, whose strength we identify from an independent source of revenue variation.

## 2.3 Setup

On the production side, each farm  $i$  produces physical output  $Y_{it}$  from labour and capital:

$$Y_{it} = e^{z_i} (L_i^{\text{fix}} + L_{it}^{\text{var}})^{\alpha_L} K_{it}^{\alpha_K}, \quad (1)$$

where  $L_{it}^{\text{var}}$  is variable labour,  $K_{it}$  is productive capital,  $L_i^{\text{fix}}$  is the farm's permanent labour endowment, and  $z_i$  is a permanent farm-specific productivity shifter. The endowment  $L_i^{\text{fix}}$  is farm-specific in the production-function definition; for tractability of the policy iteration the structural estimation in Section 5 sets  $L_i^{\text{fix}} = L^{\text{fix}}$  at the empirical median value, leaving cross-farm productivity heterogeneity to enter through  $z_i$ . Productive capital  $K_{it}$  enters production but is not the focus of the counterfactual analysis. We do not consider  $K_{it}$  with its own adjustment friction in the dynamic programme below, and the cross-farm dispersion in capital intensity is absorbed into  $z_i$ .

The farm's revenue combines physical output, the farm-year wine price, and a common time-varying revenue-scale shifter,

$$S_{it} = A_t p_{it} Y_{it}, \quad (2)$$

where  $p_{it}$  is the farm-year wine price and  $A_t$  is a multiplicative aggregate revenue shifter capturing common time-varying demand variation. Under downward-sloping aggregate demand for wine, a positive shock raises  $A_t$  commonly across farms. We treat  $A_t$  as the demand shock in the firm's optimisation.

We model the adjustment cost on variable labour as asymmetric. Expanding the workforce is costless while contracting it carries a per-worker cost. Formally,

$$C^L = c_f^L \cdot \max(L_{-1}^{\text{var}} - L^{\text{var}'}, 0), \quad (3)$$

where  $c_f^L$  is the per-worker contraction cost and the max operator selects only the contraction case ( $L_{-1}^{\text{var}} > L^{\text{var}'}$ ). The motivation for this asymmetric form is developed in the previous subsection.

Concerning the dynamic optimisation, each year the farm inherits the previous-period variable-labour stock  $L_{i,t-1}^{\text{var}}$ , observes the current demand shock  $A_t$  and capital state  $K_{it}$  given farm-specific characteristics  $(L_i^{\text{fix}}, z_i)$ , and chooses next-period variable labour  $L^{\text{var}'}$  to maximise

$$V(L_{-1}^{\text{var}}, A, K; L^{\text{fix}}, z) = \max_{L^{\text{var}'} \geq 0} \left\{ \pi - C^L + \beta_d \mathbb{E}_t V(L^{\text{var}'}, A', K'; L^{\text{fix}}, z) \right\}, \quad (4)$$

where we treat  $p_{it}$  as an idiosyncratic fluctuation conditional on  $A_t$  with  $E[p_{it} | A_t] = 1$ , so that the firm's expected revenue conditional on the state equals  $A_t Y_{it}$  and the flow operating profit is  $\pi = A Y - w L^{\text{var}'}$ , with  $Y$  given by Equation (1) evaluated at the chosen  $L^{\text{var}'}$ . The demand shock  $A_t$  and the capital state  $K_t$  follow independent stationary AR(1) processes in logs, both calibrated directly from FADN panel dynamics (Table 2). Capital is exogenous to the firm's annual choice (the model imposes no friction on the  $K$  margin), so the only choice variable is

$L^{\text{var}}$ . Throughout the paper, a stock written without subscript or prime is current-period,  $x_{-1}$  denotes the previous period,  $x'$  the next-period choice, and  $x^*$  the frictionless static optimum.

## 2.4 Identification target

The model's demand shock  $A_t$  is not observed directly. To construct its empirical counterpart, we start from farm revenue  $S_{it} = A_t p_{it} Y_{it}$  (Equation (2)). Taking logs and rearranging,  $\ln Y_{it} = \ln S_{it} - \ln A_t - \ln p_{it}$ . The within-farm-year innovation in  $\ln Y_{it}$ , which we denote  $a_{it}$ , is the residual of  $\ln S_{it}$  after farm and year fixed effects,

$$a_{it} = \ln S_{it} - \alpha_i - \lambda_t.$$

The year fixed effect  $\lambda_t$  absorbs the common time shock  $\ln A_t$ , and the farm fixed effect  $\alpha_i$  absorbs the permanent productivity  $z_i$  that enters  $\ln Y_{it}$ . Farm-year price variation  $\ln p_{it}$  that the two-way fixed effects do not absorb falls into  $a_{it}$ . This is consistent with the demand interpretation, since the IV identifies the demand-driven component of  $a_{it}$  regardless of whether that variation operates through prices or quantities. What remains in  $a_{it}$  is the farm-year variation in demand that drives within-farm input adjustment. The IV strategy in Section 3 then instruments  $a_{it}$  with weather-driven demand shocks at producer regions outside the French labour market, breaking the simultaneity between  $a_{it}$  and  $\ln L_{it}^{\text{var}}$  in Equation (5) below.

A first-order expansion of the interior policy function implied by Equation (4) delivers

$$\Delta \ln L_{it}^{\text{var}} = \beta_L a_{it} + \eta_{it}^L. \quad (5)$$

The slope  $\beta_L$  is defined as the population OLS projection coefficient of  $\Delta \ln L_{it}^{\text{var}}$  on  $a_{it}$  on the interior-margin subsample ( $L_{it}^{\text{var}} > 0$  and  $L_{i,t-1}^{\text{var}} > 0$ ), and it is the auxiliary moment that the IV in Section 3 estimates and that the indirect inference in Section 5 matches. In the frictionless static optimum implied by the first-order condition of Equation (1) on the variable-labour margin, a demand shock reaches variable labour through a leverage factor

$$1 + \frac{L_i^{\text{fix}}}{L_{it}^{\text{var},*}} > 1,$$

which exceeds one because the additive permanent-labour input does not adjust, so the variable margin alone bears the response. The full frictionless regression slope  $\beta_L^{\text{free}}$  combines this leverage factor with the production-function multiplier and the standardisation of the demand proxy; its closed form, evaluated at the central calibration point, and the simulator's intensive-margin-filtered counterpart



that Section 4.1 compares to  $\hat{\beta}_L$  are derived in [Online Appendix F](#). The contraction friction  $c_f^L$  (see Equation 3) damps the policy-function slope  $\beta_L$  below its frictionless value. A larger friction means a smaller within-farm response. The residual  $\eta_{it}^L$  bundles three distinct objects, namely the structural curvature of the policy function in the state  $L_{-1}^{\text{var}}$ , the kink at the corner  $L^{\text{var}} = 0$ , and unmodelled idiosyncratic measurement error.

The friction-damped slope itself has no closed form because the contraction cost is asymmetric; we recover  $c_f^L$  numerically by indirect inference in Section 5, matching  $\hat{\beta}_L$  together with the share of farm-years at the corner  $L^{\text{var}} = 0$ . In this section we also present the family-substitution corner, the heterogeneity in  $z$ , the calibration of the productivity distribution, and the estimation procedure. In the following section, we develop an IV strategy to estimate  $\hat{\beta}_L$  from the data.

### 3 Empirical strategy

We recover how French wine farms adjust variable labour to demand shocks. We first describe the FADN panel, then develop the instrumental-variable design that identifies this response.

#### 3.1 Data

We use farm-level data from the FADN over 2003–2024, restricted to specialised wine farms. The unfiltered panel contains 26,796 farm-year observations on 3,872 farms. After standard restrictions detailed in [Appendix A](#) and the intensive-margin filter, the headline IV sample is 15,223 farm-year observations on 1,988 farms. [Appendix A](#) reports descriptive statistics on the unfiltered panel (Table 5). All reduced-form estimates use the FADN survey weights to target the population-representative responsiveness of variable labour.<sup>4</sup>

**Labour margins.** Total labour  $L_{it}$  is measured in annual work units (UTA). Our focal margin is variable labour, defined as  $L^{\text{var}} = L - L^{\text{fix}}$ , the residual after netting out permanent labour  $L^{\text{fix}}$  (family workers and full-time employees on open-ended contracts).  $L^{\text{var}}$  captures all non-permanent hired labour. It combines fixed-term contracts (typically renewed across years to retain trained workers and returning

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<sup>4</sup>The structural simulator is an unweighted homogeneous-farm panel; under the identifying assumption that the adjustment-cost primitive  $c_f^L$  is common across the wine-farm population, the weighted data moment and the unweighted simulator moment converge to the same structural parameter.

seasonal hires) and seasonal contracts for harvest and pruning. Both components give  $L^{\text{var}}$  meaningful year-to-year persistence; it is not a pure annual flow.

**Capital, sales, climate, and trade.** Productive capital  $K_{it}^{\text{prod}}$  aggregates machinery, permanent plantations, farm buildings, and inventory, deflated by the INSEE agricultural-investment price index. Farm sales  $S_{it}$  are total wine revenue deflated by the INSEE agricultural-output price index. Weather variation enters the analysis through the instrumental-variable design of Section 3.2 via four climate dimensions at California (the wine-growing core of the United States): drought severity, spring diurnal temperature range, ripening-window mean temperature, and spring mean temperature. All four are constructed from the CRU TS 4.09 monthly climate grids of the Climatic Research Unit, University of East Anglia (Harris et al., 2020). International wine trade flows come from CEPII’s BACI bilateral trade database (HS6 product 220421), complemented by 2007 French regional wine-export tabulations from French customs (Douanes Françaises). Construction details and source files appear in [Appendix A](#).

We measure the year-to-year change in variable labour by the one-period log-difference

$$g_{it}^{\text{var}} = \ln L_{it}^{\text{var}} - \ln L_{i,t-1}^{\text{var}},$$

defined when  $L_{it}^{\text{var}}$  and  $L_{i,t-1}^{\text{var}}$  are both strictly positive and the lag is the immediately preceding year, restricting the sample to the intensive-margin subsample. Extensive-margin switches (a farm operating in year  $t - 1$  but not  $t$ , or vice versa) are outside the smooth adjustment-cost model presented in Section 5.1 and estimated in Section 5.2; we document their incidence in [Appendix A](#). Outsourcing, the substantive labour-vs-workforce comparator, is reported as an outcome of the demand shock in [Appendix B](#) on PPML levels.

### 3.2 Identification: U.S. wine-supply shocks routed through the 2007 export network

We estimate the structural slope  $\beta_L$  defined in Equation 5 by the auxiliary regression

$$g_{it}^{\text{var}} = \alpha_i + \lambda_t + \beta_L a_{it} + \xi \ln K_{i,t-1}^{\text{prod}} + u_{it}, \quad (6)$$

where  $g_{it}^{\text{var}}$  is the year-on-year log-difference of variable labour on the intensive-margin subsample,  $a_{it}$  is the standardised log-revenue innovation defined in Section 2, and  $\ln K_{i,t-1}^{\text{prod}}$  is a pre-determined scale control. The IV-identified coefficient  $\hat{\beta}_L$  is the structural moment of the model.

The shock  $a_{it}$  is a fixed-effects residual of log sales and could absorb supply-side disturbances at the farm. To recover the demand component, we instrument  $a_{it}$  with four region-year shocks built from weather at one external producer region (California, the wine-growing core of the United States). The United States is both the world's largest wine importer and a major producer; when its domestic supply falls, U.S. import demand rises and the conditions facing French wine in markets where France competes shift. The four instruments are pre-determined by the 2007 composition of French regional wine exports across destination markets.

**Construction.** For each of four climate dimensions  $c$  at California, we route the within-region z-score  $z_{\text{USA},t}^c$  through a two-layer aggregator built on the 2007 wine-trade network:

$$Z_{r,t}^c = \left( \sum_{m \in \mathcal{M}} \omega_{r,m}^0 \pi_{\text{USA},m}^0 \right) \cdot z_{\text{USA},t}^c, \quad (7)$$

where  $\omega_{r,m}^0$  is French region  $r$ 's 2007 wine-export share to market  $m$  ( $\mathcal{M}$  is the set of destination countries France exports wine to), and  $\pi_{\text{USA},m}^0$  is the share of country  $m$ 's 2007 wine exports going to the United States as an importer. The bracketed term  $\Omega_{r,\text{USA}}^0 = \sum_m \omega_{r,m}^0 \pi_{\text{USA},m}^0$  is a pre-determined exposure weight that varies across French regions. It captures region  $r$ 's position in the global wine-trade network around the United States, summed over destinations  $m$  where region  $r$  is active and where market  $m$  in turn supplies the U.S. wine market.

The four climate dimensions at California span distinct phenological windows, namely *drought severity* (negative ratio of growing-season precipitation to its long-run mean), *spring diurnal temperature range*, *ripening-window mean temperature*, and *spring mean temperature*. The four dimensions are weakly correlated at California and capture orthogonal mechanisms relevant to vine biology and U.S. wine production. Raw indices are z-scored within California across vintages; the first stage determines the empirical sign of each instrument.

**Economic channel.** The IV aggregator picks up French regions whose 2007 customer base is most exposed to the U.S. wine market through trade-network linkages. When California has bad weather, the U.S. wine market absorbs additional supply globally, and the conditions facing sellers in each wine-supplying country  $m$  shift in proportion to that country's own dependence on the U.S. as a destination ( $\pi_{\text{USA},m}^0$ ). French region  $r$ 's aggregate exposure to U.S. shocks accumulates these market-specific shifts, weighted by  $r$ 's 2007 customer composition ( $\omega_{r,m}^0$ ). Regions whose customer base concentrates in countries that supply heavily to the U.S. feel U.S. shocks most strongly through this network. The IV does not commit to a

single propagation mechanism — supply diversion, wholesale price clearing, and strategic re-routing each contribute in different markets — but the aggregate network exposure is measured by  $\Omega_{r,USA}^0$ . Because the four instruments vary at the (region, year) level, we cluster the IV standard errors at the region level.

**Identifying assumptions.** We make two identifying assumptions.

(i) *Exclusion through demand only.* The four  $Z$ 's affect French farm-level labour decisions only through the contemporaneous demand shock  $a_{it}$ . The assumption is plausible because California weather has no direct agronomic or labour-market link to French vineyards. The two regions share neither a growing season nor a labour pool, so the only route from a Californian climate shock to French hiring runs through the wine-demand channel the instrument is built to capture. Year fixed effects absorb any shock common to all French regions; the identifying variation is each region's differential exposure to U.S. weather through  $\Omega_{r,USA}^0$ .

(ii) *Pre-determined exposure weights.* The exposure weight  $\Omega_{r,USA}^0$  is constructed from 2007 French regional wine-export shares ( $\omega_{r,m}^0$ ) and 2007 BACI bilateral wine flows ( $\pi_{USA,m}^0$ ), both fixed before the panel window that supplies the time variation. The exposure-design assumption is that French regions' 2007 trade-network position relative to the United States is uncorrelated with their post-2007 farm-level labour-market shocks.

## 4 Results

### 4.1 The 4-IV: reduced form and 2SLS estimates

Identification rests on the two assumptions stated in Section 3.2. We estimate Equation (6) by 2SLS, instrumenting  $a_{it}$  with the four climate dimensions at California routed through the 2007 trade-network exposure weight  $\Omega_{r,USA}^0$ , on the variable-labour intensive-margin sample.

|                                                            | Variable labour<br>( $g_{it}^{\text{var}}$ ) |
|------------------------------------------------------------|----------------------------------------------|
| <i>Panel A: Reduced form (OLS, farm-clustered)</i>         |                                              |
| $\hat{\psi}_1$                                             | 0.0390***<br>(0.0096)                        |
| <i>Panel B: 2SLS (region-clustered, 4 instruments)</i>     |                                              |
| $\hat{\beta}_L$                                            | 0.2687**<br>(0.1202)                         |
| <i>Panel C: First-stage diagnostics (region-clustered)</i> |                                              |
| First-stage $F$                                            | 29.89                                        |
| First-stage $F$ (LOO min, $k = 3$ )                        | 10.16                                        |
| Hansen $J$ $p$                                             | 0.977                                        |
| Hansen $J$ $p$ (LOO min)                                   | 0.905                                        |
| AR 95% CI (asymptotic- $G$ )                               | [+0.012, +0.538]                             |
| AR 95% CI (small- $G$ )                                    | [-0.067, +0.610]                             |
| $N$                                                        | 15,223                                       |
| Farms                                                      | 1,988                                        |
| Regions                                                    | 14                                           |

**Note:** Outcome  $g_{it}^{\text{var}} = \ln L_{i,t}^{\text{var}} - \ln L_{i,t-1}^{\text{var}}$  on the variable-labour intensive-margin subsample ( $N = 15,223$  farm-years on 1,988 farms in 14 regions). Control:  $\ln K_{i,t-1}^{\text{prod}}$ . Fixed effects: farm and year. Weights: FADN survey weights. *Panel A:* OLS reduced form, farm-clustered SE. *Panel B:* 2SLS on the four-instrument stack, region-clustered SE. *Panel C* reports first-stage diagnostics: joint  $F$  on the four instruments and the leave-one-out (LOO) minimum  $F$  over the four three-instrument drops ( $k = 3$ ); Stock and Yogo (2005) 5%-maximal-relative-bias thresholds (Table 5.1) are 16.85 for  $k = 4$  and 13.91 for  $k = 3$ . The Hansen  $J$   $p$ -value is the IVGMM over-identification test ( $\text{df} = 3$ ; LOO  $\text{df} = 2$ ). The Anderson–Rubin 95% CIs are from grid inversion: the asymptotic- $G$  version against the  $\chi_4^2$  critical value, the small- $G$  version against the Andrews et al. (2019)  $F(4, G - 4)$  correction for  $G = 14$  region clusters. Standard errors in parentheses. *Significance:* \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

Table 1: Reduced form and instrumental variable identification on variable labour

The OLS reduced form in Panel A delivers a positive and significant slope: variable-labour hiring co-moves with the demand shock proxy. But  $a_{it}$  is the fixed-effects residual of log sales and conflates demand shifts with supply-side disturbances — weather damage at French vineyards, quality shocks, harvest-side fluctuations. The supply component of  $a_{it}$  adds variation without triggering the demand-side hiring response that  $\beta_L$  is built to measure, consistent with attenuation of the OLS slope toward zero. We therefore turn to 2SLS, which projects  $a_{it}$  onto the four climate dimensions at California routed through the 2007 trade-network exposure weight  $\Omega_{r,\text{USA}}^0$  and uses only the projected demand-side variation.

The 2SLS estimate identifies a positive but bounded labour response (Table 1, Panel B). The implied magnitude is that a one-standard-deviation increase in the demand shock raises variable labour by roughly a quarter of one log point at the intensive margin.

The 4-IV passes the standard battery (Table 1, Panel C). The joint first-stage  $F$  tests whether the four instruments jointly predict the demand shock  $a_{it}$  with enough strength for reliable 2SLS inference; the joint  $F$  comfortably clears the Stock and Yogo (2005) 5%-maximal-relative-bias threshold for  $k = 4$  instruments — the value above which the bias of 2SLS over OLS is at most 5% under weak instruments. The leave-one-out (LOO)  $F$  minimum — the worst case across the four three-instrument subsets one obtains by dropping each IV in turn — clears the 10%-maximal-bias threshold for  $k = 3$  but not the tighter 5%-bias threshold; the Anderson–Rubin weak-IV-robust CI below is the inference of record. The Anderson–Rubin (AR) weak-IV-robust confidence interval, which remains valid even when instruments are weak, excludes zero: the 4-IV is informative for the labour-response elasticity.

The Hansen  $J$  test asks a complementary question. With four instruments and one endogenous variable, the system is over-identified: each instrument provides a separate moment that estimates the same parameter  $\beta_L$ . If the four climate dimensions actually identified different elasticities — for example because one instrument violated the exclusion restriction — the  $J$  test would reject. In our case it does not reject for the headline 4-IV or for any of the four 3-IV LOO subsets (Online Appendix D): the four climate dimensions identify a common elasticity.

As an external check, the binding-catastrophe event-study (Section 2.2) implies a contemporaneous elasticity from the ratio of its event-year labour and revenue responses that is of the same order of magnitude as the 2SLS estimate, though the event-year labour coefficient is not itself individually significant. The two designs rest on entirely different variation, cross-region weather exposure here and farm-level damage timing there, so the agreement is a cross-check rather than a second identification: the event-study response is a local effect for the indemnified sub-population, whereas 2SLS targets the population-average elasticity the structural model uses.

## 4.2 Robustness of the 4-IV

The headline 4-IV is robust on three dimensions, reported in Online Appendix D. First, we ask whether the 4-IV estimate is driven by a small set of unusual observations or is broadly distributed across the IV sample. We decompose the joint 2SLS coefficient into the contribution of each (instrument, region, year) panel cell: a concentrated distribution would mean the estimate is fragile to dropping a few

cells. The normalised Herfindahl on the absolute contributions is small and no single observation drives the headline. Second, we ask whether the joint 4-IV estimate is robust to dropping any single instrument, or whether one of the four climate dimensions is anchoring the result. We re-estimate the model on each of the four three-instrument leave-one-out (LOO) subsets. A material shift in  $\hat{\beta}_L$  across the LOO drops would mean the over-identifying restrictions are not holding. The LOO point estimates remain inside the Anderson–Rubin confidence interval and the Hansen  $J$  test does not reject in any subset. Third, we ask whether 2SLS is contaminated by many-weak-instrument bias toward the OLS coefficient. We compare 2SLS to LIML, which is median-unbiased under many-weak-IV. If weak-IV bias were pulling 2SLS toward OLS, LIML would shift *away* from OLS. The LIML and 2SLS point estimates lie within rounding of each other and away from the OLS reduced form. As a separate efficiency check, optimal-weighting GMM is also close to 2SLS, indicating that the choice of weighting matrix is not driving the headline.

## 5 Structural estimation and counterfactuals

We close the model and recover the cost of the friction by indirect inference. We then quantify the operating-profit loss it creates and how that loss falls across farm types.

### 5.1 Family substitution and heterogeneity

The production function (Equation 1), the asymmetric contraction costs, and the Bellman (Equation 4) are introduced in Section 2. Two additional features of the model are central to the counterfactual analysis.

**Family-substitution corner.** The permanent labour input  $L_i^{\text{fix}}$  enters the labour aggregator additively: output remains positive when  $L_{it}^{\text{var}} = 0$ , since the permanent workforce alone produces  $e^{z_i} A_t (L_i^{\text{fix}})^{\alpha_L} K_{it}^{\alpha_K}$ . When variable-labour hire goes to zero in a low-demand year, family and full-time staff continue producing, and the corner  $L_{it}^{\text{var}} = 0$  is a viable interior choice rather than a degenerate exit. Re-entry from the corner is free. At  $L_{-1}^{\text{var}} = 0$ , the contraction term becomes  $\max(0 - L^{\text{var}'}, 0) = 0$  for any  $L^{\text{var}'} \geq 0$ . The contraction cost is paid only for moves toward zero.

**Heterogeneity.** The model generates heterogeneous incidence of the friction across farms. Low- $z$  farms operate at the corner  $L^{\text{var}} = 0$  regardless of friction (the friction

does not bind for them); intermediate- $z$  farms cross the corner across the support of  $A_t$  and are stuck paying wages they would otherwise save in low-demand years (the family-substitution-buffer channel); high- $z$  farms always hire variable labour and the friction binds on intensive-margin scaling. Section 5.3 quantifies this incidence.

## 5.2 Indirect inference and structural estimates

We estimate the adjustment-cost parameter  $c_f^L$  by indirect inference (Smith, 1993; Gourieroux and Monfort, 1996): we choose the value of  $c_f^L$  that makes the simulated panel reproduce two IV-identified data moments, the linear demand elasticity  $\hat{\beta}_L$  and the share of farm-years at the corner  $L^{\text{var}} = 0$ . The non-cost parameters of the model are calibrated outside the inference loop.

**Calibration.** The model has eleven parameters: ten calibrated outside the inference loop (Table 2) and the labour adjustment-cost parameter  $c_f^L$  estimated by indirect inference. The two shock processes and the cross-farm dispersion  $\sigma_z = 0.64$  are estimated on the FADN panel; the wage rate is calibrated to scale matching at the central productivity realisation. The capital process is fitted by a survey-weighted AR(1) on the within-farm component of  $\ln K_{it}^{\text{prod}}$ . The demand process is fitted by a survey-weighted AR(1) on the within-farm residual of  $\ln S_{it}$  after partialling out  $\ln K_{it}^{\text{prod}}$ , so that the persistence and innovation variance assigned to  $A_t$  are net of within-farm capital movements and the two AR(1) processes can be treated as independent in the simulator.<sup>5</sup>

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<sup>5</sup>The simulator uses  $\alpha_L = 0.43$ , calibrated jointly with the wage rate  $w$  so that the no-friction static optimum at the central productivity realisation matches the empirical mean of total labour. The corresponding panel-direct survey-weighted OLS estimate of  $\ln Y$  on  $\ln L$  is  $\hat{\alpha}_L = 0.355$  (SE 0.028; Table 2); the OLS estimate is subject to the standard production-function-estimation simultaneity bias from contemporaneous input choices (Olley and Pakes, 1996; Akerberg et al., 2015), and 0.43 lies within the range of agricultural production-function estimates that correct for it. The structural estimate  $\hat{c}_f^L$  is robust to  $\alpha_L \in \{0.38, 0.48\}$  with  $w$  recalibrated from the same static-optimum moment (Online Appendix E, R3).



| Parameter                       | Value     | Description                                                                                                                                                                         |
|---------------------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Production function</i>      |           |                                                                                                                                                                                     |
| $\alpha_L$                      | 0.430     | Labour share; FADN panel OLS, farm+year FE, $\hat{\alpha}_L = 0.355$ (SE 0.028), $N = 25,752$                                                                                       |
| $\alpha_K$                      | 0.085     | Capital share; same regression, $\hat{\alpha}_K = 0.095$ (SE 0.012)                                                                                                                 |
| <i>Prices and discount</i>      |           |                                                                                                                                                                                     |
| $w$                             | 0.230     | Wage rate (simulator); calibrated so $L^{\text{total},*}(z = 0, A = 1, K = 1) \approx 3$ UTA                                                                                        |
| $\beta_d$                       | 0.95      | Annual discount factor (standard)                                                                                                                                                   |
| <i>Shock processes</i>          |           |                                                                                                                                                                                     |
| $\rho_A$                        | 0.228     | AR(1) persistence of the demand shock $\ln A_t$ ; survey-weighted AR(1) on the within-farm residual of $\ln S_{it}$ after partialling out $\ln K_{it}^{\text{prod}}$ , $N = 22,591$ |
| $\sigma_{\varepsilon,A}$        | 0.285     | AR(1) innovation SD on the $K$ -purged residual                                                                                                                                     |
| $\rho_K$                        | 0.646     | AR(1) persistence of $\ln K_{it}^{\text{prod}}$ within farm; survey-weighted, two-way FE                                                                                            |
| $\sigma_{\varepsilon,K}$        | 0.382     | AR(1) innovation SD on $\ln K_{it}^{\text{prod}}$                                                                                                                                   |
| $\sigma_z$                      | 0.638     | Cross-farm permanent productivity SD; cross-farm SD of farm fixed effects in production-function regression, $N = 25,752$                                                           |
| <i>Numerical implementation</i> |           |                                                                                                                                                                                     |
| $N_A$                           | 11        | Tauchen nodes for the demand shock $A_t$ (Tauchen, 1986)                                                                                                                            |
| $N_K$                           | 7         | Tauchen nodes for the capital state $K_t$                                                                                                                                           |
| $N_z$                           | 7         | Gauss–Hermite nodes for $z_i$ ; quadrature on $\mathcal{N}(0, \sigma_z^2)$                                                                                                          |
| $L^{\text{var}}$ grid           | 45 + zero | 45 log-spaced points in $[0.20, 50]$ plus zero for the family-substitution corner                                                                                                   |

**Note:** Production-function shares ( $\alpha_L, \alpha_K$ ) are panel-direct estimates from a survey-weighted, farm-and-year fixed-effect regression of  $\ln S_{it}$  on  $\ln L_{it}$  and  $\ln K_{it}^{\text{prod}}$  on the FADN wine-farm panel 2003–2024. The simulator uses the rounded values shown in the Value column. The demand shock  $A_t$  and the capital state  $K_t$  follow independent AR(1) processes; the  $A$ -process is identified off the within-farm residual of  $\ln S_{it}$  after partialling out  $\ln K_{it}^{\text{prod}}$  so that the persistence and innovation variance assigned to  $A_t$  are net of within-farm capital movements. Capital  $K_t$  is exogenous to the firm’s choice (no  $c_f^K$  friction) but enters production through  $K^{\alpha_K}$  and is correlated within firm. The wage rate  $w$  is calibrated to scale matching: at the central productivity realisation ( $z = 0, A = 1, K = 1$ ) the static optimum matches the empirical mean of total labour ( $\sim 3$  UTA). The cross-farm productivity shifter  $z_i$  is drawn once per farm from  $\mathcal{N}(0, \sigma_z^2)$  and discretised with a 7-point Gauss–Hermite quadrature; for each  $z$ -node the Bellman is solved separately on the joint  $(L, A, K)$  state. Robustness to alternative calibrations of  $(\alpha_L, \sigma_z)$  is reported in [Online Appendix E](#).

Table 2: Calibration parameters

**Estimator.** Two data moments anchor the estimation: the IV-identified linear demand elasticity  $\hat{\beta}_L$  from Table 1 Panel B, and the share of farm-years with  $L^{\text{var}} = 0$ ,  $\text{zero}_{\text{data}} = 0.234$ . The simulator counterpart of each moment is computed on a 1,500-farm panel simulated for 30 years (with a 5-year burn-in). We choose  $c_f^L$  to minimise an absolute-distance criterion

$$D(c_f^L) = |\hat{\beta}_L - \hat{\beta}_L^{\text{sim}}(c_f^L)| + 2 |\text{zero}_{\text{data}} - \text{zero}_{\text{sim}}(c_f^L)|. \quad (8)$$

The two-to-one weighting reflects the relative magnitudes of the two moments. We use an absolute-distance criterion rather than an SE-weighted Hansen  $J$  because the simulator's zero-rate moment carries discretisation noise from the finite  $L$  grid; an inverse-variance criterion would up-weight this noise inappropriately. With two moments and one parameter, the model is over-identified by one degree of freedom; the identifying moment is  $\hat{\beta}_L$ , and the zero rate acts as a falsification check on the calibrated heterogeneity. The standard error on  $\hat{c}_f^L$  is the sandwich formula evaluated under common random numbers (Gourieroux and Monfort, 1996); the full formula, the components of the data-moment covariance matrix, and a robustness check for finite-simulation Monte-Carlo noise are reported in [Online Appendix E](#).

| Parameter                                                                                        | Estimate  |
|--------------------------------------------------------------------------------------------------|-----------|
| <i>Adjustment-cost primitive (estimated by indirect inference)</i>                               |           |
| $\hat{c}_f^L$ (per-worker labour contraction cost)                                               | 0.1200*** |
| sandwich SE                                                                                      | (0.0182)  |
| $t$ -ratio                                                                                       | +6.59     |
| <i>Fit (data target <math>\Psi_d</math> vs. simulated moment <math>\Psi_s</math> at optimum)</i> |           |
| $\hat{\beta}_L$ (data)                                                                           | +0.2685   |
| $\hat{\beta}_L^{\text{sim}}$                                                                     | +0.2592   |
| Zero-rate (data)                                                                                 | 0.234     |
| Zero-rate (sim)                                                                                  | 0.280     |

**Note:** Indirect-inference estimate of the per-worker labour contraction cost  $c_f^L$  on the labour-only Bellman of Section 5.1, using two data moments: the IV-identified linear demand elasticity  $\hat{\beta}_L$  from Table 1 Panel B and the share of farm-years at the corner  $L^{\text{var}} = 0$  on the variable-labour intensive-margin subsample (FADN 2003–2024,  $N = 15,223$  farm-years on 1,988 farms in 14 regions). Simulator setup and shock-process calibration in Table 2; criterion in Equation (8). The reported sandwich standard error is computed under common random numbers (Gourieroux and Monfort, 1996); the full formula, the components of the data-moment covariance matrix, and a finite-simulation robustness check are in Online Appendix E. Robustness to alternative state-space grids, region leave-one-out, and labour-share perturbations is also reported there. *Significance:* \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

Table 3: Indirect-inference estimate of  $c_f^L$

**Estimates.** The point estimate of  $c_f^L$  (Table 3) is statistically distinguishable from zero at conventional levels and implies a per-worker contraction cost on the order of half the wage rate (with  $w = 0.230$  from Table 2). The simulator’s elasticity is within rounding of the IV target. The zero-rate moment is matched less tightly: the simulator predicts that about 28% of farm-years hit zero variable labour against 23% in the data, a 5-percentage-point over-prediction of the corner. This overshoot inflates the threshold-crossing component of the welfare gain (Section 5.4) — farms whose ideal labour level hovers near zero — but does not affect the commercial-farms component, which operates well above the corner. The aggregate welfare estimate is therefore an upper bound on the threshold-crossing channel (Section 5.4). The estimate is robust to dropping any single region and to perturbations of the labour share with the wage rate recalibrated from the static optimum (Online Appendix E). The friction-bound direction is consistent with the natural-experiment evidence at multi-year horizon reported in Section 2.2.

### 5.3 Aggregate operating-profit gain

We compute the per-farm operating-profit gain from setting  $c_f^L = 0$  while holding the calibrated productivity-shock and cross-farm  $z$  distributions, the production-function parameters, and the wage rate at their estimated values. For each  $z$ -type we re-solve Equation (4) at  $c_f^L = 0$  and simulate the resulting policy on the same shock realisations used at the estimated  $c_f^L$ ; the within-pair difference in operating profit is the per-draw gain, averaged within  $z$  and aggregated across  $z$  using the Gauss–Hermite weights matched to  $\sigma_z = 0.64$ .

We read the gain as the friction’s effect on the average response of operating profit, not as a recoverable deadweight loss. With this framing, removing the labour adjustment friction raises operating profit by approximately 2% per farm-year on average, equivalent to about €3,300 per farm-year and an annual sector-level magnitude on the order of €140 million (about 1.4% of total sector revenue).<sup>6</sup>

Removing the labour friction raises the share of farm-years operating without variable-labour hire by approximately twelve percentage points, indicating that the asymmetric contraction friction prevents farms from contracting all the way to the family-substitution corner in low-revenue years. Without the friction, more farms drop variable-labour hire when permanent staff alone is sufficient to produce. The friction’s direct incidence is therefore on farms whose static optimum crosses the corner — they keep paying wages they would otherwise save.

The exercise is partial equilibrium: in full equilibrium, removing the friction would put upward pressure on the wage rate in high-demand years, partially offsetting the per-farm operating-profit gain. The €140M estimate is therefore an upper bound on the realisable sector-level gain. The channel decomposition is the subject of the next subsection.

### 5.4 Heterogeneous incidence

The aggregate 2% figure conceals a sharply heterogeneous distribution across farm types. The heterogeneity reflects the cross-farm productivity dispersion  $\sigma_z$  calibrated from the FADN panel; it is not separately identified by the indirect inference. Table 4 reports the per- $z$ -type operating-profit gain in absolute and population-weighted terms.

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<sup>6</sup>Model units are converted to euros via the identity  $Y = \pi + wL^{\text{var}}$ , with the conversion factor  $\kappa$  calibrated so that population-weighted simulated revenue matches the weighted mean FADN revenue of €236,000 per farm-year ( $\kappa = \text{€}71,000$  per model unit). The sector aggregate scales by the weighted population of approximately 42,300 wine farms. Discounted at 5% over a 25-year horizon, the present value is approximately €2.0 billion; treated as a perpetuity, approximately €2.8 billion.

| $z$       | Pop.<br>weight | Per-farm<br>gain (%) | Per-farm<br>gain (€/yr) | Annual sector<br>(€M) | Share<br>(%) |
|-----------|----------------|----------------------|-------------------------|-----------------------|--------------|
| −2.40     | 0.001          | +0.00%               | €0                      | 0.0                   | 0.0%         |
| −1.51     | 0.031          | +0.00%               | €0                      | 0.0                   | 0.0%         |
| −0.74     | 0.240          | +0.08%               | €40                     | 0.4                   | 0.3%         |
| +0.00     | 0.457          | +1.90%               | €1,997                  | 38.7                  | 27.3%        |
| +0.74     | 0.240          | +2.65%               | €7,891                  | 80.2                  | 56.7%        |
| +1.51     | 0.031          | +1.65%               | €17,120                 | 22.3                  | 15.7%        |
| +2.40     | 0.001          | +0.00%               | €42                     | 0.0                   | 0.0%         |
| Aggregate | 1.000          | +2.02%               | €3,344                  | 141.6                 | 100%         |

**Note:** Per- $z$  operating-profit gain from removing the labour adjustment friction  $c_f^L$ , computed by paired-draws simulation as in Section 5.3. Each row is one  $z$ -quadrature node from the calibrated  $\mathcal{N}(0, \sigma_z^2)$  with  $\sigma_z = 0.64$ ; quadrature weights sum to one across rows. *Per-farm gain* is the within- $z$  average operating-profit difference between  $c_f^L = 0$  and the estimated value, in % of the with-friction baseline and in euros. *Annual sector aggregate* scales the per-farm euro gain by the row's population weight applied to the empirical population of approximately 42,300 wine farms; the right-most column sums to the €140M aggregate reported in Section 5.3.

Table 4: Per-farm-year operating-profit gain from removing the labour friction, by productivity type

Three regimes appear in the table, in line with the model prediction of Section 5.1.

**Mechanically passive farms** ( $z \leq -0.7$ ). These farms make up about a quarter of the population; the per-farm gain is negligible (under one percent of operating profit) and their combined contribution to the sector aggregate is well below one percent. They operate at the corner  $L^{\text{var}} = 0$  regardless of the friction.

**Threshold-crossing farms** ( $z \approx 0$ , the median). These farms account for about 46% of the farm population by mass and the per-farm operating-profit gain is on the order of one percent. These are farms whose static optimum crosses the family-substitution corner across the support of  $A_t$ : in good years they hire, in bad years they should let variable labour drop to zero and let family labour absorb production. The contraction cost makes the swing too costly, so they are stuck paying wages they would otherwise save. The friction's incidence on this group is the family-substitution-buffer-disabled channel. Aggregated, this group accounts for roughly a quarter of the sector operating-profit loss.

**Above-threshold commercial farms** ( $z \geq 0.7$ ). These farms account for about 27% of the population with a much larger per-farm operating-profit gain than the median, on the order of one to two percent of operating profit per year. These are farms whose static optimum always exceeds the family-substitution threshold so they always hire variable labour. The friction binds on intensive-margin scaling: without it they would smoothly track the demand shock; with it they under-shoot in high-demand years and over-shoot in low-demand years. The per-farm euro magnitude is large because their operations are large, so each percentage point of damped scaling translates into a substantial euro figure. Aggregated, this group accounts for roughly three-quarters of the sector operating-profit loss.

## 6 Conclusion

Wine represents only 3 percent of France’s agricultural land but consumes 20 percent of the country’s pesticides, with treatment intensity nearly three times that of the average French crop.<sup>7</sup> Reducing this footprint, which is the central goal of the 2008 *Ecophyto* plan and its successive revisions, rests on substituting labour-intensive practices for chemistry — mechanical weeding, manual canopy management, and the cover-crop work that organic and high-environmental-value certifications require. The friction we estimate is not itself the constraint on these tasks, but any substitution from chemistry toward labour-intensive practices would have to flow through the variable-labour margin we do study, and we show that this margin is contractually flexible but behaviourally rigid — a friction that reshapes farm operating profit across the productivity distribution in markedly heterogeneous ways.

We use a shift-share design with four instruments, routing California weather shocks through the 2007 trade-network exposure of French regions to the US wine market. The design identifies the population-average elasticity of variable labour to a demand shock. The IV-identified response is bounded well below the frictionless static benchmark. Indirect inference on the policy slope and on the share of farm-years at the family-substitution corner places the implied per-worker contraction cost at about half the wage rate. Under the calibrated dynamic programme, removing the friction would raise sector operating profit by about 2% per year, on the order of €140 million annually.

Our estimate (€140M) is best read as an upper bound on the friction’s effect on the average operating-profit response. Three natural extensions would sharpen and enrich this magnitude. A general-equilibrium treatment with endogenous

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<sup>7</sup>Simonovici et al. (2019); the 3-percent / 20-percent contrast is also reported in Bonnefoy (2012).

wages would let friction relaxation feed back through the variable-labour market, partially absorbing the per-farm gain into wage adjustment. A re-entry friction at the family-substitution corner — search and contracting costs we currently set to zero — would generate a richer two-friction structure on the variable-labour margin. And jointly modelling outsourcing alongside variable labour would discipline a flexible margin that the data show responds positively to demand variation.

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## A Variable definitions and descriptive statistics

**Sample restrictions.** The unfiltered panel includes all FADN farms classified under the specialised-viticulture orientation. Two filters define the estimation sample. We drop singleton farms and farm-year pairs with a legal-form change between  $t - 1$  and  $t$  (which break the labour-stock continuity).

Total labour  $L_{it}$  is the farm’s annual labour input in work units (UTA), partitioned into permanent labour  $L^{\text{fix}}$  (the farm operator, family members, and full-time employees on open-ended contracts) and variable labour  $L^{\text{var}} = L - L^{\text{fix}}$  (the residual, capturing all non-permanent hired labour, namely fixed-term contracts often renewed across years and seasonal contracts for harvest and pruning, a mix carrying meaningful year-to-year worker-farm continuity). Outsourcing  $T_{it}$  is total expenditure on custom-work services purchased from agricultural contractors, deflated by the annual labour-cost index. Plant-protection expenditure and expenditure on fertilizers and soil amendments are deflated with the INSEE agricultural-input price index. Productive capital  $K_{it}^{\text{prod}}$  aggregates machinery, permanent plantations (vineyards), farm buildings, and inventory, deflated with the INSEE agricultural-investment price index; land value and forest plantations are excluded. Total farm sales  $S_{it}$  are deflated with the INSEE agricultural-output price index. The idiosyncratic revenue shock  $a_{it}$  is the survey-weighted two-way (farm and year) fixed-effects residual of  $\ln S_{it}$ , standardised to zero mean and unit variance. Replication-level field-name documentation is in the replication package’s codebook; this paper uses only the constructed variables defined here.

|                                         | All farms |         |
|-----------------------------------------|-----------|---------|
|                                         | Mean      | SD      |
| Total labour (UTA)                      | 2.7       | 2.4     |
| Permanent labour (UTA)                  | 2.1       | 1.8     |
| Variable labour (UTA)                   | 0.6       | 1.1     |
| Sales (€, deflated)                     | 237,691   | 282,985 |
| Plant protection (€, deflated)          | 9,264     | 11,022  |
| Fertilizer + amendments (€, deflated)   | 4,039     | 7,254   |
| Capital (€, deflated)                   | 261,023   | 370,804 |
| Outsourcing to agricultural contractors | 10,080    | 19,329  |
| Observations                            | 26,796    |         |
| Farms                                   | 3,872     |         |

**Note:** Means and standard deviations on the unfiltered FADN wine-farm panel, 2003–2024, survey-weighted. Labour stocks (total, permanent, variable) are in annual work units (*Unité de Travail Annuel*, UTA). Sales, plant protection, fertilizer, capital, and outsourcing are in 2010-deflated euros (sales with the INSEE agricultural-output price index; capital with the INSEE agricultural-investment price index; the remaining expenses with the INSEE agricultural-input price index). Sample sizes are the unfiltered panel (26,796 farm-years, 3,872 farms); the estimation samples on each margin are smaller after the intensive-margin and outlier filters defined in this appendix.

Table 5: Descriptive statistics (survey-weighted)

## B Outsourcing as the labour-vs-workforce comparator

The introduction’s work-versus-workforce framing rests on a direct comparison between in-house variable labour and outsourced contractor services. Agricultural contractors (*entreprises de travaux agricoles*) perform the same physical operations as the farm’s own variable labour (pruning, harvest, treatment campaigns) but transact each engagement as a one-shot purchase rather than within a continuing employment relationship. Nguyen et al. (2022) document that six in ten French farmers now use custom-work services, up from a marginal practice in the early 2000s. How the outsourcing margin co-moves with farm activity is *a priori* ambiguous: Dupraz and Latruffe (2015) find that hired labour complements both family and contract work on French field-crop farms, while contract and family labour are substitutes; Foster and Rosenzweig (2022) and Caunedo and Kala (2021) emphasise the substitution channel by which outsourced services replace owned inputs when indivisibilities or hiring frictions bind. The wine-farm reduced form below favours the complementarity reading.

The outsourcing level (see [Appendix A](#) for the field definition) departs from

the OLS-and-2SLS log-difference framework used on the labour stocks for two reasons. First, it exhibits substantial extensive-margin churn: 16.0% of farm-year observations are zero, and year-to-year transitions between zero and positive use account for over 11% of all consecutive-year pairs — transitions that the log-difference cannot accommodate by construction. Second, the intensive-margin log-differences are heavy-tailed (more than 4% of observations change by more than a factor of nine year-on-year). We therefore estimate the response by Poisson maximum likelihood with two-way fixed effects (Santos Silva and Tenreyro, 2006), which is consistent under arbitrary conditional variance and handles the extensive-margin zeros directly. Outsourcing expenditure is deflated by the annual labour-cost index rather than by the agricultural-output price index, because vineyard services are overwhelmingly labour-intensive and the labour-cost deflator measures real services purchased.<sup>8</sup>

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<sup>8</sup>The output-price deflator would mechanically compress real outsourcing expenditure in years with positive wine-price shocks and bias the estimated demand response.

|                                                                                | Outsourcing            |
|--------------------------------------------------------------------------------|------------------------|
| <i>Panel A: First stage on <math>a_{it}</math> (region-clustered)</i>          |                        |
| USA drought severity ( $Z_{r,t-1}^{\text{dry,USA}}$ )                          | −0.9476 (0.7263)       |
| USA spring diurnal range ( $Z_{r,t-1}^{\text{dtr,USA}}$ )                      | 1.3903 (1.6851)        |
| USA ripening temp ( $Z_{r,t-1}^{\text{rip,USA}}$ )                             | −2.1853 (1.8479)       |
| USA spring temp ( $Z_{r,t-1}^{\text{spr,USA}}$ )                               | −2.2587* (1.2173)      |
| Joint F                                                                        | 3.84                   |
| <i>Panel B: PPML reduced form on <math>E[T_{it}]</math> (region-clustered)</i> |                        |
| $\psi_1$ (linear, $a_{it}$ )                                                   | +0.1132***<br>(0.0082) |
| $\psi_2$ (quadratic, $a_{it}^2$ )                                              | +0.0061**<br>(0.0024)  |
| N                                                                              | 21,225                 |
| Farms                                                                          | 2,555                  |
| Regions                                                                        | 15                     |
| Frac. zero                                                                     | 16.1%                  |

**Note:** *Panel A* reports the linear first stage of  $a_{it}$  on the four headline instruments of Table 1 (the four climate dimensions at California routed through  $\Omega_{r,\text{USA}}^0$ ), region-clustered, on the PPML sample. *Panel B* reports the PPML semi-elasticity of expected outsourcing expenditure with respect to  $a_{it}$ , with farm and year fixed effects and region-clustered standard errors in parentheses. Sample is  $N \approx 20,000$  farm-years on  $\sim 2,500$  farms in 14 regions, including the extensive-margin zeros that PPML accommodates. *Significance:* \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

Table 6: Outsourcing: 4-IV first stage and PPML reduced form

The outsourcing semi-elasticity is positive at the 1% level (Panel B): a one-standard-deviation positive demand shock raises expected outsourcing expenditure by approximately eleven percent, of the same order of magnitude as the variable-labour response in the main IV table. We deliberately do not include outsourcing on the right-hand side of the variable-labour regression of Section 3.2. It is itself an outcome of the demand shock that drives labour, and adding a treatment-responsive variable as a control would introduce post-treatment bias (Angrist and Pischke, 2009). The lagged-capital control we retain is predetermined relative to the contemporaneous shock and is not subject to the same concern.

## C Robustness of the catastrophe event-study

This appendix reports two exhibits on the catastrophe event-study of Section 2.2. Table 7 is the numerical companion to Figure 1 in the body. Table 8 is the robustness

battery on the indemnity-amount cohort.

|                                                                                                        | $\ln L^{\text{var}}$<br>(1) | $\ln S$<br>(2)         |
|--------------------------------------------------------------------------------------------------------|-----------------------------|------------------------|
| <i>Panel A: Event-time profile</i>                                                                     |                             |                        |
| $h = -3$                                                                                               | 0.0188<br>(0.0426)          | -0.0229<br>(0.0146)    |
| $h = -2$                                                                                               | 0.0259<br>(0.0342)          | -0.0233<br>(0.0156)    |
| $h = -1$                                                                                               | 0.0000<br>(0.0000)          | 0.0000<br>(0.0000)     |
| $h = +0$                                                                                               | -0.0248<br>(0.0323)         | -0.0917***<br>(0.0143) |
| $h = +1$                                                                                               | -0.0405<br>(0.0422)         | -0.0976***<br>(0.0175) |
| $h = +2$                                                                                               | -0.0816<br>(0.0524)         | -0.0852***<br>(0.0204) |
| $h = +3$                                                                                               | -0.1611***<br>(0.0547)      | -0.0789***<br>(0.0220) |
| Overall ATT                                                                                            | -0.0653*<br>(0.0341)        | -0.0896***<br>(0.0135) |
| <i>Panel B: Wald-ratio elasticity <math>\hat{\theta}_L^h / \hat{\theta}_S^h</math> (within-sample)</i> |                             |                        |
| $h = +0$                                                                                               |                             | +0.27                  |
| $h = +1$                                                                                               |                             | +0.41                  |
| $h = +2$                                                                                               |                             | +0.96                  |
| $h = +3$                                                                                               |                             | +2.04                  |
| Treated farms                                                                                          |                             | 918                    |

**Note:** Sun–Abraham interaction-weighted (IW) event-study estimator (Sun and Abraham, 2021) on  $\ln L^{\text{var}}$  and  $\ln S$ , with  $h = -1$  as the omitted reference period; farm-clustered standard errors in parentheses. The treated cohort is the year of largest indemnity amount within each farm, restricted to farms with maximum indemnity  $\geq \text{€}10,000$ . Panel B reports the within-sample Wald-ratio elasticity  $\hat{\theta}_L^h / \hat{\theta}_S^h$ . Numerical companion to Figure 1 in the body. Significance: \*\*\* 1%, \*\* 5%, \* 10%.

Table 7: Event-study on the indemnity-amount cohort

**Robustness of the indemnity-amount cohort.** Table 8 reports four robustness exhibits on the indemnity-amount cohort. Panel A varies the indemnity threshold across  $\text{€}10,000$ ,  $\text{€}20,000$ ,  $\text{€}30,000$ , and  $\text{€}50,000$ , shrinking the cohort from 918 to 272 farms. The impact-year ( $h = 0$ ) coefficient is small and statistically insignificant at every threshold; the post-event labour response concentrates at the third horizon, where it is significantly negative at the two looser thresholds ( $-0.16$  at  $\text{€}10,000$ ,  $-0.15$  at  $\text{€}20,000$ ) and attenuates toward insignificance as the cohort

shrinks, consistent with the power loss from the smaller high-indemnity samples. Panel B re-estimates with standard errors clustered at the region level rather than the farm. With only 15 region clusters, conventional cluster-robust normal approximations are anti-conservative and the reported  $t$ -statistics should be referred to a  $t_{G-1}$  distribution.<sup>9</sup> Under that small- $G$  adjustment, the  $h = 3$  coefficient remains significant. Panel C compares the pooled control set with a never-indemnified-only control set. The  $h = 3$  response attenuates from  $-0.16$  to  $-0.10$  with the smaller never-only control sample, consistent with the power loss from dropping approximately 1,900 farms. Panel D reports the sum-of-squared- $t$  joint pre-trend test on  $h \in \{-3, -2\}$  under independence between event-time coefficients, which does not reject parallel trends ( $\chi^2(2) = 0.77, p = 0.68$ ). The test is conservative under positive correlation between the two pre-period coefficients, the typical case for jointly estimated event-time aggregates.

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<sup>9</sup>We use  $t_{G-1}$  critical values rather than the wild-cluster bootstrap because the number of clusters (15) is below the regime in which the bootstrap reliably controls size (MacKinnon and Webb, 2018).

|                                                                                                                                                                                                  | $h = -3$          | $h = -2$          | $h = 0$           | $h = +1$          | $h = +2$          | $h = +3$             |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------|-------------------|-------------------|-------------------|----------------------|
| <i>Panel A: Indemnity-amount threshold sensitivity (cohort = year of largest indemnity within each farm)</i>                                                                                     |                   |                   |                   |                   |                   |                      |
| INDAS $\geq$ €10,000 ( $N = 918$ )                                                                                                                                                               | 0.019<br>(0.043)  | 0.026<br>(0.034)  | -0.025<br>(0.032) | -0.041<br>(0.042) | -0.082<br>(0.052) | -0.161***<br>(0.055) |
| INDAS $\geq$ €20,000 ( $N = 619$ )                                                                                                                                                               | -0.016<br>(0.050) | -0.013<br>(0.042) | -0.005<br>(0.038) | 0.022<br>(0.050)  | -0.006<br>(0.059) | -0.146**<br>(0.066)  |
| INDAS $\geq$ €30,000 ( $N = 451$ )                                                                                                                                                               | -0.049<br>(0.056) | -0.026<br>(0.047) | -0.002<br>(0.044) | 0.049<br>(0.059)  | 0.031<br>(0.066)  | -0.108<br>(0.072)    |
| INDAS $\geq$ €50,000 ( $N = 272$ )                                                                                                                                                               | -0.042<br>(0.068) | -0.007<br>(0.055) | 0.004<br>(0.055)  | 0.061<br>(0.073)  | 0.078<br>(0.089)  | -0.067<br>(0.092)    |
| <i>Panel B: Region-clustered SE (<math>G = 15</math> regions; significance at <math>t_{G-1}</math> critical values)</i>                                                                          |                   |                   |                   |                   |                   |                      |
| REGIO-clustered                                                                                                                                                                                  | 0.019<br>(0.056)  | 0.026<br>(0.051)  | -0.025<br>(0.044) | -0.041<br>(0.058) | -0.082<br>(0.050) | -0.161***<br>(0.054) |
| <i>Panel C: Control-group robustness (broad cohort, INDAS <math>\geq</math> €10,000, IDNUM-clustered)</i>                                                                                        |                   |                   |                   |                   |                   |                      |
| Pooled (headline) ( $N_{\text{treated}} = 918$ )                                                                                                                                                 | 0.019<br>(0.043)  | 0.026<br>(0.034)  | -0.025<br>(0.032) | -0.041<br>(0.042) | -0.082<br>(0.052) | -0.161***<br>(0.055) |
| Never-indemnified controls only ( $N_{\text{treated}} = 918$ )                                                                                                                                   | -0.004<br>(0.048) | 0.008<br>(0.036)  | -0.013<br>(0.035) | -0.022<br>(0.049) | -0.043<br>(0.061) | -0.102<br>(0.065)    |
| <i>Panel D: Joint pre-trend test on <math>h \in \{-3, -2\}</math>, sum-of-squared-<math>t</math> under independence (IDNUM-clustered): <math>\chi^2(2) = 0.77</math>, <math>p = 0.682</math></i> |                   |                   |                   |                   |                   |                      |

**Note:** Sun-Abraham interaction-weighted (IW) event-study estimator (Sun and Abraham, 2021) on  $\ln L^{\text{var}}$ . Robustness exhibits on the indemnity-amount cohort of Section 2.2. Panel A varies the indemnity threshold (€10,000 to €50,000); Panel B uses region-clustered SEs with stars referred to a  $t_{G-1=14}$  distribution to account for the small cluster count; Panel C compares the headline pooled control set with never-indemnified-only controls; Panel D reports a sum-of-squared- $t$  joint pre-trend test on  $h \in \{-3, -2\}$  under independence between event-time coefficients (conservative under positive correlation, which is the typical case for jointly estimated coefficients). The reference period  $h = -1$  is omitted. Significance: \*\*\* 1%, \*\* 5%, \* 10%.

Table 8: Robustness of the event-study on the indemnity-amount cohort



## Online Appendix

### D Robustness of the IV (Online Appendix, Not for Publication)

We report three exhibits that probe the structure of the four-instrument identification in Section 3.2: a panel-cell decomposition of the headline 2SLS estimator, the drop-one-instrument leave-one-out (LOO) of the headline 4-IV stack on each input outcome, and limited-information maximum likelihood and efficient generalised method of moments alongside 2SLS on the same stack. All exhibits use the same endogenous variable ( $a_{it}$ ), the same control ( $\ln K_{i,t-1}^{\text{prod}}$ ), the same farm and year fixed effects, the same FADN survey weights, and the same region-clustered standard errors as Table 1.

**Joint cell-level decomposition.** We decompose the 4-IV 2SLS estimator at the panel-cell level via a two-step exact identity. The first step is the Imbens and Angrist (1994) weighted-average representation

$$\hat{\beta}_{2SLS} = \sum_k w_k \hat{\beta}_{IV}^k, \quad w_k = \frac{\hat{\Pi}_k Z'_k \tilde{a}}{\sum_j \hat{\Pi}_j Z'_j \tilde{a}},$$

of the joint estimator as a weighted average of just-identified single-IV estimators, with weights  $w_k$  determined by the joint first-stage projection (the  $\hat{\Pi}_k$  are the joint first-stage coefficients on  $Z_k$  and  $\tilde{a}$  is the demand shock residualised on the controls and fixed effects of Equation (6)). The second step decomposes each  $\hat{\beta}_{IV}^k$  at the panel-cell level by writing  $\alpha_{r,t}^k = (Z_k \tilde{a})|_{r,t} / \sum_{r',t'} (Z_k \tilde{a})|_{r',t'}$  for the share of  $Z'_k \tilde{a}$  contributed by panel cell  $(r, t)$ . Composing the two steps gives the joint cell weight on  $(k, r, t)$  as  $w_k \alpha_{r,t}^k$ , and Table 9 reports the resulting decomposition.<sup>10</sup>

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<sup>10</sup>This is a regression-sample decomposition of the joint 2SLS estimator: it identifies which (instrument, region, year) cells of the regression sample drive each instrument's contribution to the headline estimator. Joint cell weights  $w_k \alpha_{r,t}^k$  can be negative, which is mechanical for shift-share constructions and does not signal a failure of the over-identifying restrictions.

| <i>Panel A: IA94 weights and per-instrument concentration</i>                                      |        |                      |                  |                      |
|----------------------------------------------------------------------------------------------------|--------|----------------------|------------------|----------------------|
| Instrument                                                                                         | $w_k$  | $\hat{\beta}_{IV}^k$ | Herfindahl       | % cells $\alpha < 0$ |
| USA dry sev.                                                                                       | 0.014  | 0.166                | 1.317            | 49.7%                |
| USA dtr spr.                                                                                       | 0.404  | 0.240                | 0.117            | 41.0%                |
| USA ripen tmp                                                                                      | 0.246  | 0.327                | 0.202            | 45.0%                |
| USA spring tmp                                                                                     | 0.336  | 0.265                | 0.125            | 45.3%                |
| $\hat{\beta}_{2SLS} = \sum_k w_k \hat{\beta}_{IV}^k = 0.2685$ (decomposition error = $7.8e - 16$ ) |        |                      |                  |                      |
| <i>Panel B: Joint 4-IV cell distribution (1,288 (instrument,r,t) cells)</i>                        |        |                      |                  |                      |
| Joint Herfindahl on $ w_k \alpha_k  / \sum  w_k \alpha_k $                                         |        |                      |                  | 0.0038               |
| % cells with $w_k \alpha_k < 0$                                                                    |        |                      |                  | 45.3%                |
| Top-10 share of $\sum  w_k \alpha_k $                                                              |        |                      |                  | 11.2%                |
| Largest single cell share                                                                          |        |                      |                  | 1.62%                |
| <i>Panel C: Top-10 cells by <math> w_k \alpha_k </math></i>                                        |        |                      |                  |                      |
| Instrument                                                                                         | Region | Year                 | $\alpha_{r,t}^k$ | $w_k \alpha_{r,t}^k$ |
| USA ripen tmp                                                                                      | 54     | 2021                 | 0.228            | 0.056                |
| USA dtr spr.                                                                                       | 54     | 2004                 | 0.134            | 0.054                |
| USA dtr spr.                                                                                       | 54     | 2022                 | 0.123            | 0.050                |
| USA spring tmp                                                                                     | 54     | 2004                 | 0.112            | 0.038                |
| USA ripen tmp                                                                                      | 54     | 2023                 | 0.135            | 0.033                |
| USA spring tmp                                                                                     | 54     | 2005                 | -0.098           | -0.033               |
| USA dtr spr.                                                                                       | 72     | 2002                 | 0.080            | 0.032                |
| USA ripen tmp                                                                                      | 72     | 2023                 | -0.129           | -0.032               |
| USA ripen tmp                                                                                      | 21     | 2021                 | 0.122            | 0.030                |
| USA spring tmp                                                                                     | 54     | 2009                 | 0.088            | 0.030                |

**Note:** Two-step decomposition of the headline 4-IV 2SLS estimator. *Panel A* reports the Imbens and Angrist (1994) instrument-level weights  $w_k$  across the four climate dimensions at California and the just-identified single-IV estimators  $\hat{\beta}_{IV}^k$ , with the within-instrument Herfindahl on the panel-cell  $\alpha_{r,t}^k$  weights. *Panel B* summarises the joint cell distribution: the Herfindahl on the normalised  $|w_k \alpha_{r,t}^k|$  distribution, the share of total  $|w_k \alpha^k|$  accounted for by the top-10 cells, and the largest single cell's share. *Panel C* lists the top-10 cells by  $|w_k \alpha^k|$ . Same outcome, endogenous variable, control, fixed effects, and weights as Table 1.

Table 9: Joint cell-level decomposition of the 4-IV 2SLS estimator on  $g^{\text{var}}$

Two features of the decomposition are worth flagging. Panel B confirms that the joint 2SLS is broadly distributed across the (instrument, region, year) cells of the regression sample, so no small set of observations drives the headline. Panel A reports the IA94 instrument-level weights  $w_k$  across the four climate dimensions at California (drought severity, spring diurnal range, ripening-window temperature, spring temperature); the three temperature- and diurnal-range indices carry

the bulk of the IA94 weight, while the growing-season-drought index carries only about 1%, so identification effectively rests on three instruments, with drought-severity retained because it remains a valid moment under the over-identification test.

**Drop-one-IV LOO.** Table 10 reports the headline 4-IV in the first row and the four 3-IV variants that drop each instrument in turn, on the same  $L^{\text{var}}$  intensive-margin sample as Table 1. Two patterns matter for the interpretation. First, the Hansen  $J$  over-identification test does not reject in any of the four drop-one rows. The four climate dimensions identify the same demand-elasticity object whichever three-instrument subset one looks at. Second, the headline 4-IV Anderson–Rubin 95% CI excludes zero under the asymptotic  $\chi_k^2$  critical value, and the four instruments together deliver stronger identification than any three-instrument subset.

| Specification       | $k$    | $F$   | $\hat{\beta}$ (SE) | AR $\chi^2$ 95% CI | AR $F$ 95% CI    | $J$ $p$ |
|---------------------|--------|-------|--------------------|--------------------|------------------|---------|
| Headline 4-IV       | 4      | 29.90 | +0.2685** (0.1202) | [+0.012, +0.537]   | [−0.067, +0.610] | 0.977   |
| Drop USA dry sev.   | 3      | 37.39 | +0.2690** (0.1252) | [+0.039, +0.511]   | [−0.011, +0.561] | 0.905   |
| Drop USA dtr spr.   | 3      | 10.16 | +0.2842** (0.1330) | [+0.034, +0.562]   | [−0.013, +0.655] | 0.955   |
| Drop USA ripen tmp  | 3      | 31.02 | +0.2491** (0.0980) | [+0.025, +0.542]   | [−0.034, +0.606] | 0.978   |
| Drop USA spring tmp | 3      | 36.21 | +0.2604** (0.1289) | [−0.008, +0.516]   | [−0.060, +0.575] | 0.910   |
| $N$                 | 15,221 |       |                    |                    |                  |         |
| Farms               | 1,987  |       |                    |                    |                  |         |
| Regions             | 14     |       |                    |                    |                  |         |

**Note:** The first row reproduces the headline 4-IV. Each subsequent row drops one instrument and re-estimates the resulting 3-IV on the same panel and same specification. Region-clustered standard errors in parentheses. “AR  $\chi^2$  95% CI” is the Anderson–Rubin grid inversion against the  $\chi_k^2$  critical value (asymptotic- $G$  reference). “AR  $F$  95% CI” is the same grid inversion against the small-cluster-corrected  $k F(k, G - k)$  critical value of Andrews et al. (2019).  $J$   $p$  is the Hansen over-identification  $p$ -value computed via the efficient two-step IVGMM Sargan–Hansen statistic. *Significance:* \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

Table 10: Drop-one-IV leave-one-out and Anderson–Rubin CI on  $g^{\text{var}}$

**LIML and efficient GMM.** Table 11 reports the limited-information maximum likelihood (LIML) estimator and the two-step efficient generalised method of moments (GMM) estimator alongside 2SLS on the headline 4-IV stack. LIML is the canonical robustness against many-weak-instrument bias (Staiger and Stock, 1997; Andrews et al., 2019). Under weak-IV bias toward OLS, 2SLS would be pulled toward the OLS reduced form (Panel A of Table 1) while LIML would not. The LIML

and 2SLS point estimates lie within rounding of each other, and efficient GMM is similarly close. All three estimators agree on sign, magnitude, and significance pattern.

| Estimator | $\hat{\beta}$ | SE (DoF-corrected) |
|-----------|---------------|--------------------|
| 2SLS      | +0.2685**     | 0.1158             |
| LIML      | +0.2703**     | 0.1165             |
| GMM       | +0.2790**     | 0.1303             |
| $N$       |               | 15,221             |
| Farms     |               | 1,987              |
| Regions   |               | 14                 |

**Note:** All three estimators on the 4-IV stack with outcome  $g_{it}^{\text{var}}$  on the  $L^{\text{var}}$  intensive-margin sample. Region-clustered standard errors in parentheses. Computed on the within-transformed sample (farm and year demeaned, survey-weighted); standard errors include the absorbed-fixed-effects degrees-of-freedom correction  $\sqrt{(N - k)/(N - k - n_{\text{id}} - n_T + 1)}$ , which inflates each row's SE by approximately seven percent relative to the linearmodels default that ignores the absorbed FE. *Significance:* \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

Table 11: 2SLS, LIML, and efficient GMM on  $g^{\text{var}}$

**Alternative weather-IV stacks.** The headline stacks four climate dimensions at California (drought severity, spring diurnal range, ripening-window temperature, spring temperature) routed through the 2007 trade-network aggregator  $\Omega_{r,\text{USA}}^0$ . We report two alternative four-instrument stacks that share the drought-severity and spring-diurnal-range core with the headline but swap the other two dimensions:

- *Variant 1.* 'dry\_severity + dtr.spring + harvest\_pre + spring\_tmn\_min' — replacing the ripening- and spring-temperature dimensions with harvest-month precipitation and the spring minimum temperature.
- *Variant 2.* 'dry\_severity + dtr.spring + harvest\_pre + pet\_ripening' — replacing the spring temperature with potential evapotranspiration over the ripening window.

Both variants are routed through the same  $\Omega_{r,\text{USA}}^0$  aggregator at California. Table 12 reports the headline alongside the two variants.

|                                                                       | Headline<br>(1)       | Variant 1<br>(2)      | Variant 2<br>(3)    |
|-----------------------------------------------------------------------|-----------------------|-----------------------|---------------------|
| <i>Panel A: First stage on <math>a_{it}</math> (region-clustered)</i> |                       |                       |                     |
| Joint F (4 IVs)                                                       | 29.89                 | 25.88                 | 24.61               |
| Leave-one-out F (min over 4 drops, $k = 3$ IVs)                       | 10.16                 | 14.74                 | 15.15               |
| Hansen J $p$ (df = 3)                                                 | 0.977                 | 0.216                 | 0.265               |
| Leave-one-out Hansen J $p$ (min over 4 drops)                         | 0.905                 | 0.137                 | 0.137               |
| <i>Panel B: 2SLS (region-clustered)</i>                               |                       |                       |                     |
| $\hat{\beta}_L$                                                       | +0.2687**<br>(0.1202) | +0.1408**<br>(0.0707) | +0.2127<br>(0.1312) |
| 95% CI                                                                | [+0.03, +0.50]        | [+0.00, +0.28]        | [−0.04, +0.47]      |
| $N$                                                                   | 15,223                | 15,223                | 15,223              |
| Farms                                                                 | 1,988                 | 1,988                 | 1,988               |
| Regions                                                               | 14                    | 14                    | 14                  |

**Note:** All three columns are 4-IV stacks of climate dimensions at California, each routed through the 2007 trade-network exposure weight  $\Omega_{r,USA}^0$  on the  $L^{\text{var}}$  intensive-margin sample. *Headline:* dry\_severity, dtr\_spring, ripening\_tmp, spring\_tmp. *Variant 1:* dry\_severity, dtr\_spring, harvest\_pre, spring\_tmn\_min. *Variant 2:* dry\_severity, dtr\_spring, harvest\_pre, pet\_ripening. Same outcome, control ( $\ln K_{i,t-1}^{\text{prod}}$ ), fixed effects (farm and year), weights (FADN survey weights), and clustering (region) as Table 1. Panel A reports joint first-stage diagnostics; Panel B reports the 2SLS coefficient on the variable-labour log-difference and the asymptotic 95% confidence interval  $\hat{\beta}_L \pm 1.96 \cdot \text{SE}$ , reported for cross-spec comparability; the headline AR weak-IV-robust CI is in Table 1, Panel C. Standard errors in parentheses. *Significance:* \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.10$ .

Table 12: Alternative weather-IV stacks at California: headline vs. two variants

The point estimate on  $\beta_L$  is positive across the three constructions (Table 12, Panel B). The headline 2SLS estimate of +0.27 has a region-clustered standard error of 0.12, so its 95% confidence interval is approximately [+0.03, +0.50] (Table 12, Panel B, "95% CI" row). Both variant point estimates (+0.14 and +0.21) sit inside that interval, so the variants are not statistically distinguishable from the headline. The dispersion of point estimates across alternative climate-dimension choices, with  $\hat{\beta}_L$  ranging from +0.14 to +0.27, is within the sampling variation of the headline itself. The Hansen  $J$  over-identification test does not reject in any column. Sign, over-identification, and statistical magnitude all survive alternative choices of the four climate dimensions.

## E Structural robustness on $c_f^L$ (Online Appendix, Not for Publication)

**Sandwich standard error: formula and finite-simulation robustness.** The reported sandwich SE on  $\hat{c}_f^L$  in Table 3 is computed as  $\hat{V}_{c_f^L} = (J' \hat{V}_\Psi^{-1} J)^{-1}$ , where  $J$  is the  $2 \times 1$  Jacobian of the simulated moments with respect to  $c_f^L$  at the optimum (forward finite difference) and  $\hat{V}_\Psi$  is the diagonal data-moment covariance matrix with the cluster-robust variance of  $\hat{\beta}_L$  from Section 4.1 and the binomial variance of the zero rate. The Jacobian is evaluated under common random numbers. The same shock path is used at the optimum and at the perturbed parameter, so the simulator-induced moment variance at the simulated panel size ( $1,500 \times 30 = 45,000$  farm-years) is suppressed below the cluster-robust data-moment variance that enters  $\hat{V}_\Psi$ , and one simulated panel ( $S = 1$ ) is sufficient. The reported SE inherits the data-moment uncertainty in  $\hat{V}_\Psi$ , not Monte-Carlo simulator noise; the Gourieroux and Monfort (1996) finite-simulation inflation factor is  $(1 + 1/S)$ , so as an upper bound the SE could be inflated up to  $\sqrt{2} \approx 1.41$  to absorb the residual Monte-Carlo component, leaving the  $t$ -statistic above four and the parameter clearly distinguishable from zero.

We probe the indirect-inference fit of the labour adjustment-cost parameter along three orthogonal directions: the regional composition of the IV sample, the outlier trim on the log-difference outcome, and the production-function calibration of the labour share. Table 13 reports the perturbed data targets and the corresponding structural fits. The exhibit's purpose is the *relative stability* of  $\hat{c}_f^L$  across the three perturbation classes. To keep nineteen Bellman re-solves tractable, the table uses a coarser state-space discretisation than the headline pipeline of Table 3 (35-point log-spaced positive  $L^{\text{var}}$  grid with  $L_{\text{HI}} = 20$  UTA and 5-point Gauss-Hermite on  $z$ , against the headline's 45-point grid with  $L_{\text{HI}} = 50$  UTA and 7-point Gauss-Hermite). At the coarser grid the absolute-distance argmin lands on the same baseline value as the headline grid (Table 3); the criterion  $D(c_f^L)$  is shallow in the neighbourhood of its minimum, so the coarser grid tracks the headline argmin without displacement. Across all nineteen perturbations the appendix-grid estimate stays in a tight band around the headline value; that band is the exhibit's substantive content.

**Region leave-one-out.** The first block re-estimates the IV-identified labour elasticity on each of the fourteen 13-region subsamples, then refits  $c_f^L$  to the perturbed target. The friction stays in a tight band across the leave-one-out drops; no single

region drives the estimate.

**Trim sensitivity.** The second block re-runs the perturbation across three values of the log-difference outlier trim,  $\tau \in \{\ln 6, \ln 9, \ln 12\}$ . The friction is essentially constant across the three trims; the estimate is not sensitive to the trim cutoff.

**Production-share sensitivity.** The third block varies the labour share  $\alpha_L$  across  $\{0.38, 0.43, 0.48\}$ . For each  $\alpha_L$  we re-calibrate the wage rate from the static optimum at  $(z, A) = (0, 1)$  so that the no-friction static optimum still matches the empirical mean labour, and re-solve the value function on the headline  $c_f^L$  grid. The friction varies smoothly with  $\alpha_L$  but the qualitative magnitude — a contraction cost on the order of one tenth of the wage — holds across the three labour-share values. The cross-farm productivity dispersion  $\sigma_z$  and the AR(1) shock variance  $\sigma_\varepsilon$  are not re-estimated under the perturbed labour share; doing so is out of scope for this exhibit.

|                                                                                                  | $\hat{\beta}_L$ | $\widehat{\text{zero}}$ | $\hat{c}_f^L$ | $D$   |
|--------------------------------------------------------------------------------------------------|-----------------|-------------------------|---------------|-------|
| <i>Baseline (App. E grid)</i>                                                                    | +0.2685         | 0.234                   | 0.120         | 0.028 |
| <i>R1. Region leave-one-out</i>                                                                  |                 |                         |               |       |
| Drop REGIO=21                                                                                    | +0.3283         | 0.252                   | 0.100         | 0.059 |
| Drop REGIO=22                                                                                    | +0.2720         | 0.235                   | 0.120         | 0.030 |
| Drop REGIO=24                                                                                    | +0.2776         | 0.233                   | 0.120         | 0.040 |
| Drop REGIO=26                                                                                    | +0.2525         | 0.241                   | 0.120         | 0.024 |
| Drop REGIO=42                                                                                    | +0.1708         | 0.242                   | 0.160         | 0.017 |
| Drop REGIO=43                                                                                    | +0.2618         | 0.235                   | 0.120         | 0.025 |
| Drop REGIO=52                                                                                    | +0.2420         | 0.239                   | 0.140         | 0.035 |
| Drop REGIO=54                                                                                    | +0.1492         | 0.223                   | 0.160         | 0.044 |
| Drop REGIO=72                                                                                    | +0.2437         | 0.214                   | 0.140         | 0.086 |
| Drop REGIO=73                                                                                    | +0.2952         | 0.234                   | 0.120         | 0.056 |
| Drop REGIO=82                                                                                    | +0.2518         | 0.243                   | 0.120         | 0.019 |
| Drop REGIO=91                                                                                    | +0.2825         | 0.218                   | 0.120         | 0.075 |
| Drop REGIO=93                                                                                    | +0.2650         | 0.235                   | 0.120         | 0.023 |
| Drop REGIO=94                                                                                    | +0.2831         | 0.233                   | 0.120         | 0.045 |
| <i>R2. Trim sensitivity</i>                                                                      |                 |                         |               |       |
| $\tau = \ln 6$                                                                                   | +0.1981         | 0.234                   | 0.140         | 0.030 |
| $\tau = \ln 9$ (headline)                                                                        | +0.1985         | 0.234                   | 0.140         | 0.030 |
| $\tau = \ln 12$                                                                                  | +0.2194         | 0.234                   | 0.140         | 0.021 |
| <i>R3. Production-share sensitivity (target fixed at headline; <math>w</math> re-calibrated)</i> |                 |                         |               |       |
| $\alpha_L = 0.38$                                                                                | +0.2685         | 0.234                   | 0.100         | 0.041 |
| $\alpha_L = 0.43$ (headline)                                                                     | +0.2685         | 0.234                   | 0.120         | 0.028 |
| $\alpha_L = 0.48$                                                                                | +0.2685         | 0.234                   | 0.140         | 0.047 |

**Note:** Each row reports a perturbed data target ( $\hat{\beta}_L, \widehat{\text{zero}}$ ) and the corresponding structural fit  $\hat{c}_f^L$  that minimises the absolute-distance criterion  $|d_L| + 2|d_{\text{zero}}|$  on the simulation grid (Equation 8). R1 drops one of the fourteen IV-sample regions and recomputes the target on the remaining 13-region subsample. R2 varies the log-difference outlier trim. R3 varies the labour share with the wage rate re-calibrated from the static optimum so that the no-friction static optimum still matches the empirical mean labour;  $\sigma_z$  and  $\sigma_\varepsilon$  are not re-estimated.

Table 13: Structural robustness of  $\hat{c}_f^L$  to region LOO, trim, and labour-share perturbations

## F Frictionless-limit benchmark for $\beta_L$ (Online Appendix, Not for Publication)

Section 2.4 introduces the policy-function slope  $\beta_L$  and its frictionless counterpart  $\beta_L^{\text{free}}$ . The friction-damped slope has no closed form under the asymmetric con-



traction cost and is recovered numerically by indirect inference (Section 5.2); the frictionless slope does have one. We derive it below, then explain why Section 4.1 benchmarks  $\hat{\beta}_L$  against a simulator-based rather than the analytical frictionless slope.

**The frictionless limit is static.** At  $c_f^L = 0$  the contraction term in the Bellman (Equation (4)) vanishes, so the previous-period stock  $L_{-1}^{\text{var}}$  drops out of the problem. The farm re-optimises from scratch each year, solving

$$L_{it}^{\text{var}*} = \arg \max_{L^{\text{var}} \geq 0} e^{z_i} A_t (L_i^{\text{fix}} + L^{\text{var}})^{\alpha_L} K_{it}^{\alpha_K} - w L^{\text{var}}.$$

**The static first-order condition gives the level elasticity.** The interior first-order condition equates the marginal product of labour to the wage and solves for total labour,  $L_i^{\text{fix}} + L_{it}^{\text{var}*} = (\alpha_L e^{z_i} A_t K_{it}^{\alpha_K} / w)^{1/(1-\alpha_L)}$ . Differentiating in logs with respect to the demand shock  $\ln A_t$ , and using that  $L_i^{\text{fix}}$  is fixed so the variable margin absorbs the entire adjustment,

$$\frac{\partial \ln L_{it}^{\text{var}*}}{\partial \ln A_t} = \left(1 + \frac{L_i^{\text{fix}}}{L_{it}^{\text{var}*}}\right) \cdot \frac{1}{1 - \alpha_L}. \quad (9)$$

Two forces lift the response. The Cobb–Douglas multiplier  $1/(1 - \alpha_L)$  is the standard diminishing-returns term. A higher  $A$  raises the marginal product of labour, and total labour rises until the first-order condition re-equilibrates. The leverage factor  $1 + L_i^{\text{fix}}/L^{\text{var}*}$  is specific to our setup. Because permanent labour does not adjust, the variable component alone bears any proportional change in total labour.

**From the level elasticity to the regression slope.** The auxiliary regression (Equation (6)) projects the year-on-year difference  $\Delta \ln L^{\text{var}}$  on the standardised demand residual  $a_{it}$ , not the level  $\ln L^{\text{var}}$  on  $\ln A$ . Two conversions bridge the gap. First, differencing an AR(1) shock of persistence  $\rho$  scales the level elasticity by  $(1 - \rho)$ . Second, standardising  $a_{it}$  to unit variance scales it by  $\sigma_{\ln A}$ , and in this step the multiplier  $1/(1 - \alpha_L)$  cancels, because the residual  $a_{it}$  is itself proportional to  $1/(1 - \alpha_L)$ . Combining the two,

$$\beta_L^{\text{free}} = \underbrace{\left(1 + \frac{L_i^{\text{fix}}}{L^{\text{var}*}}\right) \cdot \frac{1}{1 - \alpha_L}}_{\text{level elasticity}} \cdot \underbrace{(1 - \rho) \sigma_{\ln A}}_{\text{difference} \times \text{standardisation}}. \quad (10)$$

With the calibrated values  $\alpha_L = 0.43$ ,  $\rho_A = 0.23$ ,  $\sigma_{\ln A} = 0.29$  (Table 2;  $\sigma_{\ln A} = \sigma_{\varepsilon, A} / \sqrt{1 - \rho_A^2}$ ),  $L^{\text{fix}} = 2.0$  UTA and  $L^{\text{var}*} = 1.0$  UTA at the central calibration point, Equation (10) gives  $\beta_L^{\text{free}} \approx 1.18$ .

**Benchmarking against the simulator.** The analytical value is evaluated at the central farm. The IV estimates  $\beta_L$  on the intensive-margin sample ( $L^{\text{var}} > 0$  in both  $t$  and  $t - 1$ ), which over-samples high-productivity farms operating well above the family-substitution corner, where the leverage factor  $1 + L^{\text{fix}}/L^{\text{var}*}$  is much smaller. Re-solving the Bellman at  $c_f^L = 0$  and running the same auxiliary regression on the same intensive-margin filter yields a frictionless-simulator slope well below 1.18. Constructed on the same sample as  $\hat{\beta}_L$ , that simulator slope is the directly comparable benchmark, and the estimated friction-damped slope is roughly half of it: the labour adjustment friction halves the policy-function slope relative to the frictionless static optimum.